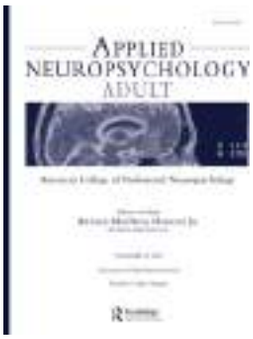

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Meta-analysis of Controlled Oral Word Association Test (COWAT) FAS performance in amnestic mild cognitive impairment and cognitively unimpaired older adults

Kacie Bauer[a] and Michael Malek-Ahmadi[b]*

[a]Division of Neurobiology, Neuroscience and Cognitive Science, University of Arizona, Tucson, AZ, USA; [b]Banner Alzheimers Institute, Phoenix, AZ, USA

ABSTRACT

Findings from several studies have shown that amnestic mild cognitive impairment (aMCI) older adults have significantly lower performance on phonemic fluency tasks relative to cognitively unimpaired (CU) older adults. These findings suggest that nonmemory domains, such as executive function, are impacted in aMCI. As Alzheimer's disease (AD) research has shifted toward identifying and characterizing preclinical AD, there is a need to identify subtle but significant cognitive changes that are below the threshold for clinical impairment. The aim of this meta-analysis was to examine phonemic fluency differences between aMCI and CU older adults. Data from 18 studies were included in this analysis that found that aMCI individuals' phonemic fluency performance was approximately seven points lower than CU individuals ($\Delta = -7.31$, 95% CI [−9.10, −5.52], $z = -8.01$, $p < 0.001$), which represents a medium effect size of ($g = 0.61$, 95% CI [0.46, 0.76], $z = 7.90$, $p < 0.001$). Normative conversion of the aMCI groups' raw scores showed that all were in the normal range of performance. The findings of this meta-analysis demonstrate that significant subclinical deficits in phonemic fluency can be present in aMCI. This should prompt greater use of phonemic fluency tasks in outcome measures for observational and intervention studies.

KEYWORDS
Executive function; phonemic fluency; preclinical Alzheimer's disease; verbal fluency

Introduction

Amnestic mild cognitive impairment (aMCI) is characterized by objective memory impairment that is at least 1.5 standard deviations below age- and education-adjusted normative performance and are at an increased risk for progressing to clinical Alzheimer's disease (AD) (Petersen & Negash, 2008). Although memory performance has been the primary focus of research relating to aMCI, there is evidence that nonmemory domains may also be impacted, suggesting that subtle, but significant, changes in these domains may be informative (Brandt & Manning, 2009; Chehrehnegar et al., 2020). Several studies have reported that aMCI patients' cognitive performance is intermediate to cognitively unimpaired (CU) participants and early-stage AD patients not only on episodic memory measures, but also on verbal fluency and executive function measures (Chehrehnegar et al., 2020; Kramer et al., 2006; Ribeiro et al., 2006). Kramer et al. (2006) suggest that isolated memory impairments are uncommon in aMCI patients; therefore, including measures of fluency and executive function in testing for aMCI may enhance the sensitivity of cognitive composite scores.

Recent evidence has suggested that executive function decline can be detected prior to episodic memory changes, further suggesting that performance in nonmemory domains is an important component of the overall cognitive changes seen in aMCI individuals (Chehrehnegar et al., 2020). This point is underscored by evidence showing that executive function measures discriminate best between aMCI individuals who go on to develop AD relative to those who do not (Economou et al., 2007). Malek-Ahmadi et al. (2011) demonstrated that aMCI individuals performed significantly worse on phonemic and category fluency measures when compared to age-, sex-, and education-matched CU older adults where large effect sizes were noted (phonemic fluency: Cohen's $d = 1.33$, category fluency: Cohen's $d = 1.05$). Although Malek-Ahmadi et al. (2011) reported large effect sizes between CU and aMCI groups, it was noted that the aMCI group's phonemic and category fluency scores were in the normal range of performance.

Phonemic fluency is often measured using the Controlled Oral Word Association Test (COWAT), where individuals are given one minute to say as many words as possible starting with a specific letter, often F, A, or S (indicated by COWAT-FAS). Other variations of this test use C, F, and L or P, R, and W. Category fluency is measured in a similar manner, except that individuals are asked to name items from a specific category (e.g., animals, fruits, etc.). A previous meta-analysis by Rodriguez-Aranda and Martinussen (2006) analyzed COWAT performance across different decades of age, from the 20s to 80s. They report that phonemic

CONTACT Michael Malek-Ahmadi michael.malekahmadi@bannerhealth.com Banner Alzheimers Institute, 901 E. Willetta St., Phoenix, AZ 85006, USA.
*Department of Biomedical Informatics, University of Arizona College of Medicine-Phoenix, Phoenix, AZ, USA.

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fluency performance begins to decline in the 60s, then rapidly declines through the 80s. Given that the those in the age range of 60–80 are at the highest risk for developing aMCI and AD, using phonemic fluency measures may be helpful in determining the likelihood of further cognitive decline and may also be important in developing sensitive efficacy measures for clinical trials. Research diagnostic criteria for prodromal AD recognizes that decline in nonmemory domains of cognition are likely to be informative in predicting cognitive decline when used with sensitive biomarkers of *in vivo* AD pathology (Sperling et al., 2011). The ability to detect subtle but significant declines in cognition is underscored by ongoing efforts toward identifying preclinical AD, which is characterized by the presence of *in vivo* blood and/or neuroimaging markers of AD pathology without clinically significant cognitive impairment (Jack et al., 2018). Despite this biologically-centered approach, there is still a need to demonstrate clinically meaningful benefits of disease-modifying therapies in clinical trials. This is particularly important where sensitivity to subclinical cognitive changes are needed. Although several studies have reported that COWAT-FAS performance is significantly lower in aMCI individuals relative to CU individuals (Brandt & Manning, 2009; Chehrehnegar et al., 2020; Malek-Ahmadi et al., 2011; Nutter-Upham et al., 2008; Putcha et al., 2020), phonemic fluency measures do not appear to be widely-used in studies of preclinical AD.

The aim of this meta-analysis is to characterize COWAT-FAS performance differences between aMCI and CU older adults. A moderate to large overall effect size would provide convincing evidence for the wider use of phonemic fluency in assessing subtle cognitive changes associated with preclinical AD.

Methods

Inclusion criteria

Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were used to conduct this meta-analysis (Liberati et al., 2009). Prior to conducting the literature searches, the following criteria for study selection and inclusion were established: (1) the data could not come from a treatment or intervention trial; (2) the study must report raw means and standard deviations for COWAT-FAS performance in both the CU and aMCI groups; and (3) the study must use Petersen criteria (Petersen & Negash, 2008) to classify its aMCI subjects.

We attempted to limit the inclusion of studies to those that used single-domain aMCI subjects; however, this proved to be untenable as the majority of included studies did not specify whether their aMCI groups were single- or multi-domain.

Literature search terms

Using PubMed, the first search term used was "verbal fluency and mild cognitive impairment," which yielded 1,398 results. Of these, 45 were selected for further screening from which nine studies were able to be included. A second search using the term "phonemic fluency and mild cognitive impairment" yielded 157 results, but all were duplicated from the first search. Additional searches using the terms "FAS and mild cognitive impairment" and "letter fluency and mild cognitive impairment" yielded 107 and 53 results, respectively; however, they did not yield any additional studies beyond those in the previous searches.

The same search terms were also entered in the PsycInfo database. The search term "letter fluency and mild cognitive impairment" yielded 93 results from which one additional study was obtained. A search of "phonemic fluency and mild cognitive impairment" in PsycInfo yielded 75 results, but all were duplicates of those studies found in previous searches. From all of the literature searches, ten studies met the selection criteria. Eight additional studies that met the inclusion criteria were found after manually reviewing the Reference sections for each of the ten studies that brought the final number of included studies to 18. The large number of studies found through the manual review reflects studies where the primary research questions were not related to verbal fluency, but COWAT-FAS data for CU and aMCI samples were still reported. The literature searches and study selection were carried out in parallel by the authors (KB and MM) and any discrepancies in study selection were discussed and resolved.

Studies using other phonemic fluency letter groupings (CFL, PRW, etc.) were excluded as they were not as commonly used as FAS, and it has been shown that FAS and CFL differ in difficulty (Barry et al., 2008) and their inclusion would result in a significant confounding of the aggregated data.

Data quality and extraction

The quality of each study was assessed using the NIH Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies (Study Quality Assessment Tools, 2021) that yields ratings of "Good," "Fair," or "Poor" based on 14 items. Of the 18 included studies, 10 were graded as "Good" and eight were graded as "Fair." From the selected studies means and standard deviations for age, education, and FAS scores were extracted for both the CU and aMCI groups, as well as the sample sizes for each group. The diagnostic method for each study was also classified (Consensus, Clinical, Algorithmic, Unspecified) as well as the study setting (Clinic, Community). It was noted that several studies utilized both clinic (aMCI) and community (CU) setting recruitment.

Statistical analysis

The primary analyses were carried out using the "meta" and "metafor" packages in R 4.0.3. Analyses of the standardized mean difference (SMD) (Hedge's *g*) and the raw mean difference (RMD) for COWAT-FAS were carried out. RMDs for age and education were also reported. Results from

random effects analyses were reported and the I^2 statistic was used to quantify between-study heterogeneity which was classified as low, moderate, or high based on proposed guidelines (Higgins et al., 2003). Additional SMD analyses were carried out for subgroups based on study setting (clinic vs. community) and diagnostic method (clinical vs. consensus). The Egger's test was used to determine the presence of publication bias among the included studies.

In order to provide results with additional clinical utility, the means of the aMCI groups' COWAT-FAS values were converted to *z*-scores using age- and education-adjusted normative data (Loonstra et al., 2001).

Results

A total of 54 studies were screened for inclusion from which 18 met criteria for inclusion (Figure 1). Reasons for exclusion included: used non-FAS letters, used only one letter for the fluency task, did not use Petersen criteria to classify aMCI. Three studies did not report means and standard deviations for FAS results. The authors of these articles were contacted in which one was able to provide the raw means and standard deviations, whereas the raw data were not available for the other two articles. A comprehensive list of the included studies with quantitative and qualitative information is shown in Table 1. From the eighteen studies, the total sample size was 2,732 (CU = 2006, aMCI = 726). All

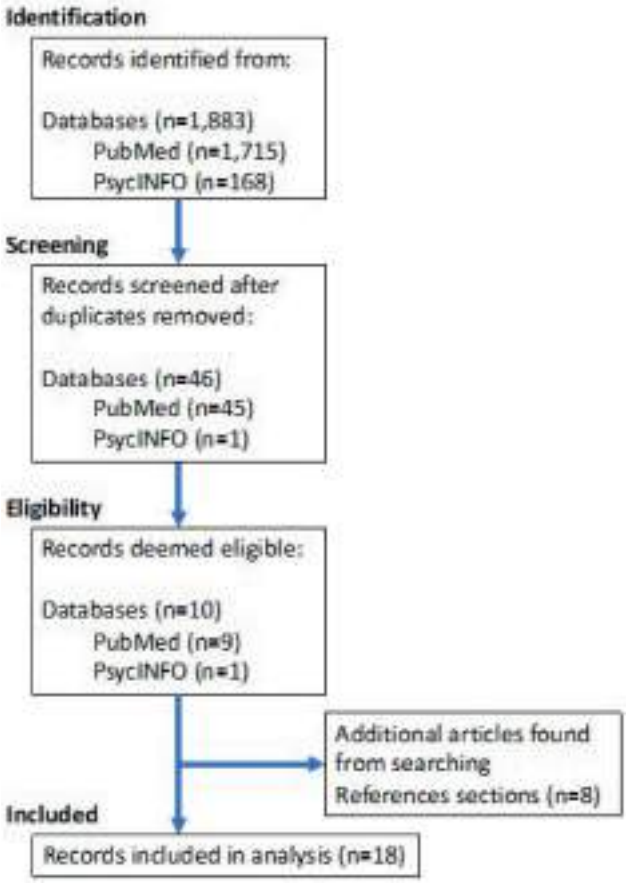

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart of study screening and selection.

Table 1. Methodologic and demographic characteristics of studies included in the meta-analysis.

Study	Setting	Diagnostic Approach	CU Sample size	CU Age	CU Education	aMCI Sample size	aMCI Age	aMCI Education
Armentano et al. (2009)	Clinic	Clinical	35	66.7 ± 7.9	8.8 ± 4.7	13	73.5 ± 7.3	9.5 ± 4.9
Behforuzi et al. (2013)	Clinic and Community	ns	22	72.9 ± 8.2	16.8 ± 3.5	11	68.2 ± 10.8	15.5 ± 3.1
Clark et al. (2014)	Clinic and Community	Consensus	25	70.1 ± 6.9	16.2 ± 2.5	23	70.7 ± 7.4	14.8 ± 2.7
Clark et al. (2009)	Community	Consensus	96	74.4 ± 8.5	14.1 ± 2.9	21	80.5 ± 7.4	14.1 ± 2.9
Demetriou and Holtzer (2017)	Community	Consensus	353	76.1 ± 6.6	14.6 ± 3.0	13	82.4 ± 6.5	13.7 ± 2.8
Eckerström et al. (2020)	Clinic and Community	ns	126	65.4 ± 7.5	12.2 ± 2.9	43	66.1 ± 6.4	11.8 ± 3.5
Holtzer et al. (2020)	Community	Consensus	454	75.6 ± 6.4	14.8 ± 2.8	60	78.1 ± 6.8	13.1 ± 3.1
Jacobs et al. (2020)	Community	Consensus	294	75.6 ± 6.3	14.8 ± 2.8	50	77.4 ± 6.2	15.3 ± 2.7
Johns et al. (2012)	Clinic	Clinical	32	71.8 ± 5.0	14.4 ± 3.2	40	72.4 ± 8.6	13.1 ± 3.1
Malek-Ahmadi et al. (2011)	Community	Consensus	33	74.8 ± 5.8	14.2 ± 2.3	33	74.8 ± 5.8	14.2 ± 2.3
McDonnell et al. (2020)	Clinic and Community	ns	99	63.5 ± 11.7	16.3 ± 2.5	90	64.8 ± 11.1	16.1 ± 2.5
Mirandez et al. (2017)	Clinic and Community	Clinical	37	72.5 ± 4.5	13.5 ± 3.4	30	74.5 ± 5.4	12.0 ± 3.1
Nutter-Upham et al. (2008)	Community	Consensus	33	71.1 ± 5.5	17.3 ± 2.4	37	71.8 ± 6.7	16.2 ± 2.9
Price et al. (2010)	Clinic	Consensus	33	75.5 ± 6.2	13.5 ± 2.4	33	77.6 ± 7.2	12.2 ± 3.1
Putcha et al. (2020)	Community	Consensus	99	68.7 ± 7.5	16.1 ± 2.6	32	70.4 ± 7.2	16.4 ± 2.7
Rinehardt et al. (2014)	Clinic and Community	Consensus	34	75.2 ± 5.3	13.7 ± 1.8	34	74.7 ± 6.0	13.8 ± 2.2
Teng et al. (2013)	Clinic and Community	Clinical	111	70.3 ± 8.0	17.0 ± 2.3	138	73.8 ± 8.2	15.6 ± 2.7
Weakley et al. (2013)	Community	ns	90	71.4 ± 9.4	15.7 ± 2.7	25	72.7 ± 10.5	16.4 ± 2.9
			63.5 (22–454)† *2006*	71.8 ± 7.1*	14.7 ± 2.8*	33 (11–138)† *726*	73.6 ± 4.6*	14.1 ± 1.9*

Notes. mean ± standard deviation; ns = not specified; †median (range); *aggregate sample size*; *pooled from raw values.

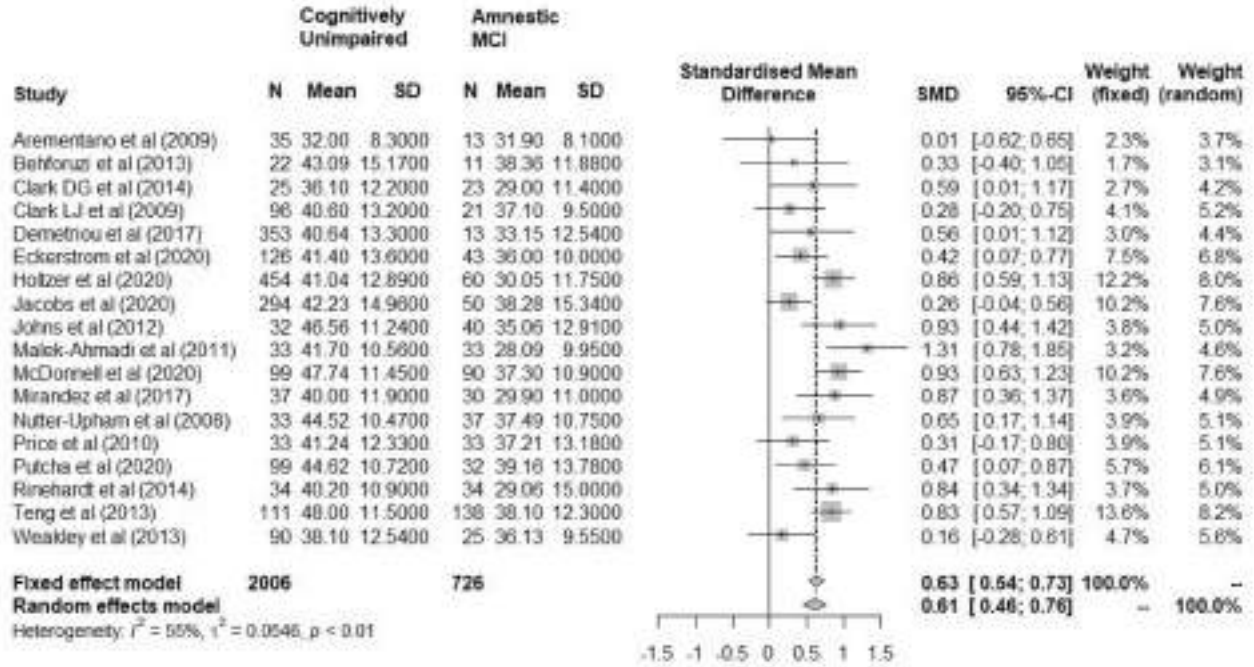

Figure 2. Forest plot of standardized mean differences for Controlled Oral Word Association Test-FAS (COWAT-FAS) differences between cognitively unimpaired and amnestic mild cognitive impairment.

of the included studies were published between 2008 and 2020.

Consensus diagnosis was the most common diagnostic method ($n = 10$); whereas, four studies used a clinical diagnosis to characterize their subjects. The remaining four studies did not explicitly state whether a consensus or clinical diagnosis was used. For the study setting, eight used community-based samples and three used clinic-based samples while the remaining seven studies recruited their CU participants from the community and their aMCI participants from clinics. The aMCI groups were approximately two years older than the CU groups ($\Delta = 1.80$, 95% CI [1.01, 2.82]) and the difference in years of education between CU and aMCI groups was negligible ($\Delta = 0.60$, 95% CI [0.16, 0.97]).

The RMD of COWAT-FAS showed that the aMCI groups' performance was significantly lower than that of the CU groups ($\Delta = -7.31$, 95% CI [−9.10, −5.52], $z = -8.01$, $p < 0.001$) which represents a medium effect size ($g = 0.61$, 95% CI [0.46, 0.76], $z = 7.90$, $p < 0.001$). There was a moderate level of between-study heterogeneity ($I^2 = 55\%$, 95% CI [24%, 74%]) A forest plot of the SMDs for each study along with the results of the random effects analysis are shown in Figure 2.

For studies that used a clinical diagnosis ($n = 4$) a larger effect size was noted when compared to the main analysis ($g = 0.72$, 95% CI [0.40, 1.05], z = 4.38, $p < 0.001$); however, the between-study heterogeneity was still moderate ($I^2 = 52\%$, 95% CI [0%, 84%]). For studies that used a consensus diagnosis ($n = 10$), the effect size was nearly identical to that of the main analysis ($g = 0.60$, 95% CI [0.40, 0.81], $z = 5.79$, $p < 0.001$) and the level of between-study heterogeneity was still moderate ($I^2 = 54\%$, 95% CI [6%, 78%]).

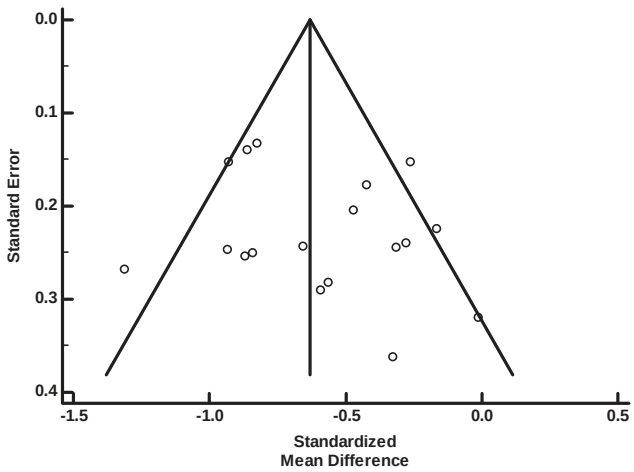

Figure 3. Funnel plot for studies included in the meta-analysis.

Among studies that utilized clinic-based samples ($n = 3$) a more modest effect size for COWAT-FAS differences was noted ($g = 0.44$, 95% CI [−0.08, 0.97], $z = 1.66$, $p = 0.10$) where heterogeneity was moderate ($I^2 = 66\%$, 95% CI [0%, 90%]), while community-based studies ($n = 8$) yielded an effect size that was more similar to the overall analysis ($g = 0.56$, 95% CI [0.31, 0.81], $z = 4.36$, $p < 0.001$) with moderate heterogeneity ($I^2 = 66\%$, 95% CI [28%, 84%]). Interestingly, the effect size for studies that utilized a combination of clinic-based and community-based recruitment was the largest ($g = 0.74$, 95% CI [0.58, 0.90], $z = 8.86$, $p < 0.001$) with low heterogeneity ($I^2 = 15\%$, 95% CI [0%, 59%]).

The Egger's test for publication bias was not statistically significant ($p = 0.33$), indicating that publication bias did not significantly influence the effect size estimates. A funnel plot of the effect size estimates is shown in Figure 3 where

Table 2. Normative conversions for aMCI COWAT-FAS performance.

Study	aMCI COWAT-FAS	*Age-adjusted normative *z*-score	*Education-adjusted normative *z*-score
Armentano et al. (2009)	31.9 ± 8.1	−0.03	0.14
Behforuzi et al. (2013)	38.4 ± 11.9	0.48	−0.22
Clark et al. (2014)	29.0 ± 11.4	−0.26	−0.98†
Clark et al. (2009)	37.1 ± 9.5	0.59	−0.33
Demetriou and Holtzer (2017)	33.2 ± 12.5	0.29	−0.65
Eckerström et al. (2020)	36.0 ± 10.0	0.29	0.45
Holtzer et al. (2020)	30.1 ± 11.8	−0.18	−0.90†
Jacobs et al. (2020)	38.3 ± 15.3	0.47	−0.23
Johns et al. (2012)	35.1 ± 12.9	0.22	−0.49
Malek-Ahmadi et al. (2011)	28.1 ± 9.9	−0.33	−1.05†
McDonnell et al. (2020)	37.3 ± 10.9	0.39	−0.31
Mirandez et al. (2017)	29.9 ± 11.0	−0.19	−0.91†
Nutter-Upham et al. (2008)	37.5 ± 10.8	0.41	−0.30
Price et al. (2010)	37.2 ± 13.2	0.39	−0.32
Putcha et al. (2020)	39.2 ± 13.8	0.54	−0.16
Rinehardt et al. (2014)	29.1 ± 15.0	−0.26	−0.98†
Teng et al. (2013)	38.1 ± 12.3	0.46	−0.25
Weakley et al. (2013)	36.1 ± 9.6	0.30	−0.40
Overall values	34.5 ± 11.7	0.20 ± 0.31	−0.44 ± 0.41

Notes. aMCI = amnestic mild cognitive impairment; COWAT-FAS = Controlled Oral Word Association Test-FAS; mean ± standard deviation; *Used normative data from Table 5 of Loonstra et al. (2001); †*z*-score falls in the Low Average range.

the plot's relative symmetry further demonstrates the negligible effect of publication bias on these results.

Age-adjusted normative conversion of the aMCI groups' COWAT-FAS scores indicated that their overall performance was in the Average range ($z = 0.20$) where the *z*-scores for all studies ranged from −0.33 to 0.59 (Table 2). Education-adjusted *z*-scores yielded a similar finding, but it is noted that five studies had aMCI groups that performed in the Low Average range.

Discussion

This meta-analysis shows that there is a meaningful subclinical difference in phonemic fluency performance between aMCI and CU older adults. This study found a 7-point difference between aMCI and CU individuals, which represents a medium effect size and bears relevance for detecting important subclinical cognitive changes that may go beyond episodic memory tests to differentiate aMCI individuals from those who are CU. Although the aMCI groups' performance is considered normal by conventional clinical standards, the medium effect size for their lower performance relative to CU older adults indicates that subtle declines in nonmemory domains may also be indicators of disease progression. Treatment studies that enroll aMCI individuals may benefit from having phonemic fluency measures incorporated with memory measures which could enhance the sensitivity of cognitive endpoints to detect treatment effects.

The findings of this study also show that the setting in which participants were recruited had a significant impact on observed effect sizes. Notably, the analysis of studies that used a combination of clinic-based and community-based recruitment yielded the lowest between-study heterogeneity and the most robust effect size when compared to studies that solely used clinic- or community-based samples. Others have noted that differences in the sources of participant recruitment can significantly impact findings of aMCI studies (Malek-Ahmadi, 2016; Petersen, 2010; Petersen et al., 2014). Studies that utilize clinic-based aMCI samples are more likely to have higher rates of progression to AD (Petersen et al., 2014) and lower rates of reversion to CU (Malek-Ahmadi, 2016). It is possible that studies which utilize clinic- and community-based recruitment may provide less-biased estimates of cognitive and clinical outcomes.

Although this meta-analysis analyzed differences between CU and aMCI individuals, its findings are prescient given the continued focus on identifying subclinical cognitive changes that occur in the presymptomatic stages of AD (Langbaum et al., 2015; Weintraub et al., 2018). Secondary prevention trials for AD are ongoing in individuals who are positive for AD-related biomarkers, but whose cognition is still intact. This conceptualization of preclinical AD (Jack et al., 2018) is now the prevailing paradigm for AD research and treatment (Langbaum et al., 2013) and is further supported by regulatory authorities who have suggested the possibility of approval for disease-modifying therapies based on a measure of cognition that shows "persuasive effects" (United States Food & Drug Administration, 2018). It is possible that asymptomatic individuals with significant levels of AD pathology manifest similar, but slightly smaller, subclinical declines in phonemic fluency as their aMCI counterparts. In this regard, the inclusion of phonemic fluency measures in cognitive endpoints for both AD prevention and aMCI clinical trials may provide additional sensitivity to cognitive outcomes (Weintraub et al., 2018). Since the aMCI groups in the included studies yielded COWAT performances that were mostly in the Average range of functioning, the clinical value of these findings is unclear. However, there is significant scientific value in a cognitive test that can detect a medium effect size as this is likely to enhance the sensitivity to change of a composite measure of cognition that includes tests from other cognitive domains.

A limitation of this meta-analysis is that it framed phonemic fluency performance as a measure of executive function as there is debate about whether verbal fluency tasks

are measures of language or executive function. Aita et al. (2019) characterize COWAT-FAS as a measure of executive function, while Whiteside et al. (2016) contend that it is a language-based measure. Rabinovici et al. (2015) propose that fluency (phonemic and semantic) is one of four primary components that underlie executive function which supports the interpretation of COWAT-FAS used in this study. It is noted that there can be significant overlap between cognitive domains (Harvey, 2019); therefore, it is difficult to classify a single cognitive test as a pure measure of one particular domain. An additional limitation to this study is the inability to differentiate single-domain aMCI from its multidomain counterpart in several of the included studies, which indicates the possibility that aMCI individuals with executive function impairment were included in the analysis. This may have resulted in reported effects sizes that are larger than what would normally be expected between CU and single-domain aMCI, thus inflating the effect sizes of this meta-analysis.

The results of this meta-analysis show that there are significant subclinical differences on COWAT-FAS performance between aMCI and CU older adults, a finding that should be given greater consideration when constructing cognitive outcomes for observational and intervention studies.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Kacie Bauer http://orcid.org/0000-0001-8580-8639
Michael Malek-Ahmadi http://orcid.org/0000-0001-9901-3650

Data availability statement

Data used in this manuscript can be requested from Dr. Michael Malek-Ahmadi via e-mail.

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